

Why do we need a skeleton?



By the end of this lesson I will be able to:

- Describe the difference between vertebrates and invertebrates.
- Name at least 10 bones of the human skeleton using their scientific names.
- State the 4 functions of the skeletal system.

Sort these animals into 2 groups of your choice.

Eagle

Worm

Snake

Wasp

Next Step: Add 2 more animals to each group

e.g.

Group 1	Group 2
Animal	Animal
Animal	Animal

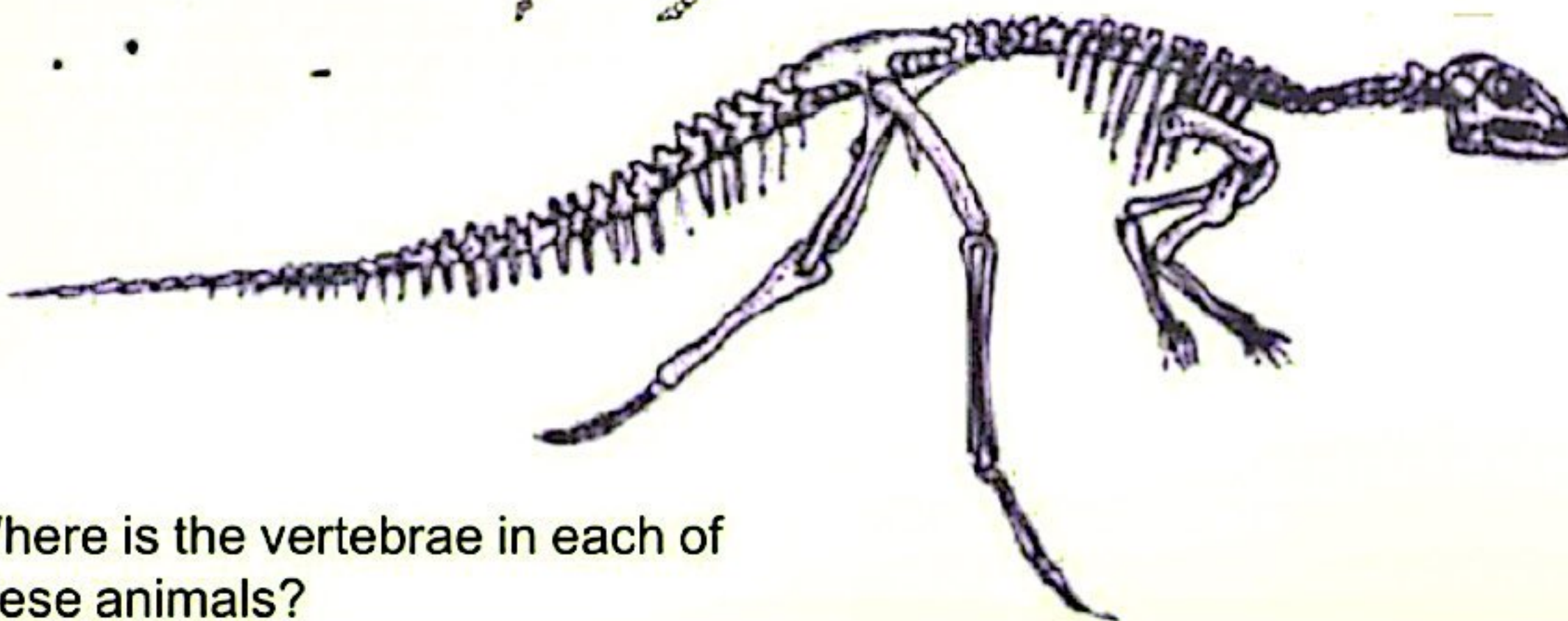
LO1: Describe the difference between vertebrates and invertebrates.



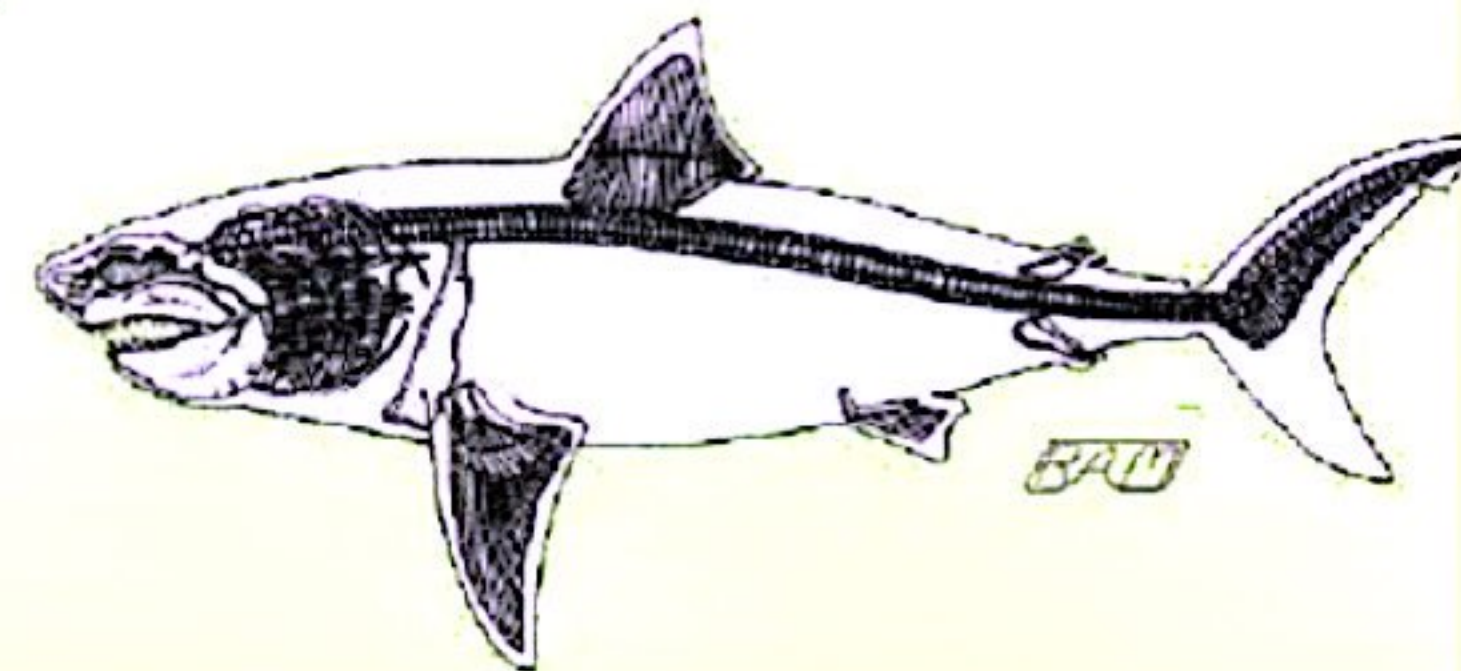
Are these all mammals?



Can we get fossils of invertebrates?



Where is the vertebrae in each of these animals?



Skeletal System

Four type of cells involved in movement

① Ciliated cells

② Amoeboid cells

③ Flagellated cells

④ Muscle cells

Three kinds of Skeletons.

① Hydrostatic skeleton

② Exoskeleton

③ Endoskeleton

Hydrostatic Skeleton.

→ Core of liquid surrounded

by tension-resistant sheath of

circular & longitudinal muscles.

contraction of muscles → push →

against hydrostatic skeleton → generates

Hydrostatic Skeleton.

→ Core of liquid surrounded by tension-resistant sheath of circular & longitudinal muscles.
contraction of muscles → push → against hydrostatic skeleton → generates body movement.

Sea-anemone.

Can be found in

1 1

gills / hydrostatic skeleton → generates
body movement.

Sea-anemone:-

Circular muscle → contract
↓

body → lengthens.

Scanned with CamScanner

Longitudinal muscles $\longrightarrow$ contract
shortens. $\longleftarrow$

Earthworms:-

- $\longrightarrow$ Alternative contraction and relaxation of longitudinal & circular muscles.
- $\longrightarrow$ Prevents from collapsing.
- Forms of hydrostatic skeleton.

circular muscles.

→ Prevents from collapsing.

Forms of hydrostatic skeleton.

Gastrovascular cavity → Flatworms

Rhynchocoel → Nemertea

Pseudocoelom → Aschelminthes

Coelom → Annelids

Hemocoel → Molluscs.

Exoskeleton



Coelom → Annelids

Hemocoel → Molluscs.

Exoskeleton



Encloses the body.

→ points for muscle attachment.

→ Support

→ Protection

Epidermis → Cuticle → waterproofing
↓

→ Encloses the body.

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↓
Chitin + Protein

Crustaceans → CaCO_3 → rigid.

Shedding of exoskeleton → periodically
Cuticle → Thin / flexible regions
by joints.

Muscles → antagonistically → Movement.
Joints → elastic proteins

↓
Resilin

↓
Store energy on compression

Cuticle → Thin / flexible regions
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↓
Resilin

↓
→ Store energy on compression
→ release for movement.

Endoskeleton

Endoskeleton

→ Body tissues encloses it.

→ Sponges:-

↳ spicules + Spongin fibers.

→ Adult sponges attach to substrate →
No need for muscles.

→ Echinoderms:-

→ Composed of small,
calcareous plates → ossicles.

→ CaCO_3 .

ENDOSKELETONS

SPONGES



ECHINODERMATA

